Bihar Engineering University, Patna

B.Tech 5th Semester Exam., 2021

Course: B.Tech | *Time: 3 Hours*
Code: 104502 | *Subject: DIGITAL SIGNAL PROCESSING* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 **Answer any seven from the following :** **[2 x 7 = 14]**

(a) Show that energy of the signal and its DFT are equal.
(b) State the time-shifting property of DFT.
(c) Is DCT a linear transform? Justify.
(d) Name the properties of DTFT.
(e) What is the difference between decimation in time and decimation in frequency?
(f) State the properties of ROC.
(g) Name at least two applications of chirp-z transform.
(h) Name one advantage and one disadvantage of the window technique for designing FIR filter.
(i) What is Twiddle factor?
(j) In the direct computation of the N-point DFT of $x(n)$, how many complex additions and complex multiplications can be obtained?

Q.2 **Solve both questions :** **[8+6=14]**

(a) Compute circular convolution using DFT and IDFT for the given sequences $x_1(n) = \{2, 3, 1, 1\}$ and $x_2(n) = \{1, 3, 5, 3\}$.
(b) Give the steps for obtaining tabular and functional convolution sum with suitable examples.

Q.3 **Solve both questions :** **[7 x 2 = 14]**

(a) Determine the zero-input response of the system described by homogeneous second-order difference equation $y(n) - 3y(n-1) - 4y(n-2) = 0$.
(b) Check whether the given systems are linear, shift variant, causal and stable :
(i) $y[n] = x[4n + 1]$
(ii) $y[n] = x[n]u[n]$

Q.4 **Solve both questions :** **[7 x 2 = 14]**

(a) Compute the convolution $y(n) = x(n) * h(n)$ of the following pair of given signals :
(i) $x(n) = a^n u(n)$, $h(n) = b^n u(n)$ when $a \neq b$ and when $a = b$.
(ii) $x(n) = 1$ for $n = -2, 0, 1; 2$ for $n = -1; 0$ elsewhere
$h(n) = \delta(n) - \delta(n-1) + \delta(n-4) + \delta(n-5)$
(b) Obtain the 8 point DIF and DIT FFT of given sequence {8, 8, 8, 0, 1, 4, 2, 3}.

Q.5 **Solve both questions :** **[7 x 2 = 14]**

(a) Determine the region of convergence of right-sided, left-sided, and finite-duration two-folded sequences.
(b) Using long division, determine the inverse z-transform of $x(z) = (1 + 2z^{-1})/(1 - 2z^{-1} + z^{-2})$, if
(i) $x(n)$ is causal.
(ii) $x(n)$ is anti-causal.

Q.6 **Solve both questions :** **[9+5=14]**

(a) Determine the 8-point DFT of the signal $x(n) = \{1, 1, 1, 1, 1, 1, 0, 0\}$ and also draw the magnitude and phase.

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(b) State at least three properties of DFT.

Q.7 **Solve both questions :** **[8+6=14]**

(a) Compute the 16-point DFT of the sequence $x(n) = \cos(\pi/2)n$, $0 \le n \le 15$, using the radix-4 decimation-in-time algorithm.

(b) Explain the architecture of TMS processor with suitable block diagram.

Q.8 **Solve both questions :** **[7 x 2 = 14]**

(a) Develop a radix-2 DIF FFT algorithm for $N = 8$ and draw the signal flow graph.

(b) Obtain direct form-I, direct form-II and cascade structures for the following system :
$y(n) = 3/4\ y(n-1) - 1/8\ y(n-2) + x(n) + 1/3\ x(n-1)$

Q.9 **Write short notes on the following :** **[7+7=14]**

(a) Goertzel algorithm

(b) Linear phase filter using Hamming window

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